Mobile Game A/B Test Analysis — Cookie Cats

A SQL-driven evaluation of a progression-gate experiment on 90,189 mobile-game users

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Context

Cookie Cats is a popular mobile puzzle game that uses a progression "gate" to pace player engagement. In the live experiment analyzed here, 90,189 newly installed users were randomly assigned to one of two variants — gate_30 (control, gate at level 30) or gate_40 (treatment, gate at level 40). The product hypothesis was that pushing the gate later would let players experience more content before friction, improving retention and overall game-round engagement.

This case study quantifies the actual effect using SQL — covering allocation balance, headline retention and engagement metrics, two-proportion z-tests with 95% confidence intervals, engagement-segment shifts, and a 1,000-run bootstrap robustness check. The recommendation at the end is a binary rollout decision with supporting evidence.

Methodology

All analyses run on a local DuckDB instance against the raw CSV. The two-proportion z-test uses the pooled-proportion variance estimator for the null-hypothesis test, and the unpooled estimator for the confidence interval on the lift — standard practice. Significance is reported at the 90%, 95%, and 99% levels. The bootstrap is implemented in pure SQL using a CROSS JOIN against GENERATE_SERIES(1, 1000) with a RANDOM() < 0.8 row filter to produce 1,000 ~80% subsamples; the empirical 95% interval is the [2.5, 97.5] quantile of the resampled lift distribution.

Allocation balance was verified before any outcome analysis: gate_30 received 49.56% of users and gate_40 received 50.44%, well within tolerance for random assignment, so no sample ratio mismatch invalidates the results.

Headline Results

Metric	Control %	Treatment %	Abs. lift (pp)	z-stat	Significance
Day 1 retention	44.82	44.23	−0.59	−1.78	p ≈ 0.07
Day 7 retention	19.02	18.20	**−0.82**	**−3.16**	**p < 0.01**

The Day 7 effect is the central finding. The treatment arm shows a 0.82-percentage-point absolute drop in 7-day retention (−4.31% relative), with a z-statistic of −3.16 — significant at the 99% confidence level. The 95% confidence interval on the lift is [−1.33pp, −0.31pp], entirely below zero. The Day 1 effect is in the same direction (−0.59pp) but only weakly significant (p ≈ 0.07) with a 95% CI that just crosses zero. The negative effect grows in magnitude and statistical strength as the time horizon extends — consistent with the hypothesis that the gate change weakens the early "hook" that drives habit formation, with the consequence surfacing over time rather than immediately.

Engagement Segment Shift

Engagement segment	gate_30 (% of arm)	gate_40 (% of arm)	Shift (pp)
Never played (0 rounds)	4.33	4.52	+0.19
Low (1–10 rounds)	35.20	35.74	+0.54
Medium (11–50 rounds)	35.55	34.07	**−1.48**
High (51–200 rounds)	19.09	19.91	+0.82
Power (200+ rounds)	5.83	5.76	−0.07

Decomposing the population by total game rounds played in the 14-day observation window reveals where the retention drop comes from. The treatment arm loses 1.48 percentage points from the medium-engagement segment (11–50 rounds) and gains users in the never-played and low-engagement segments. This is the texture behind the average effect: the late gate does not improve outcomes for committed players, it converts what would have been medium-engagement players into shallow players who churn early. The hypothesis was that removing early friction would let more players "get to the good stuff" — instead, removing early friction appears to remove the early commitment signal that turns trial users into engaged ones.

Bootstrap Robustness Check

To check whether the Day 7 result is robust to outliers or quirks in the sample, the analysis re-ran the lift calculation 1,000 times, each on an ~80% random subsample of users. The results closely match the analytical z-test: mean lift of −0.82 percentage points with a standard deviation of 0.13pp; empirical 95% interval [−1.08pp, −0.56pp]. Critically, 0% of the 1,000 resamples produced a positive lift — the negative direction of the effect is essentially unanimous across resamples. The analytical and empirical confidence intervals overlap tightly, giving high confidence that the finding is real rather than driven by a few outlier users or a fragile statistical assumption.

Recommendation

Do not roll out the gate_40 variant. Keep the gate at level 30. The treatment is statistically significantly worse on Day 7 retention with high confidence (99% analytical, 100% empirical), and the engagement-segment shift suggests the mechanism is loss of early habit formation — not a measurement artefact or a sample issue.

A natural follow-up experiment, if the product team wants to revisit the gate position, would be to test an earlier gate (e.g., level 20) against the current level 30, isolating the direction of the hook hypothesis. A separate orthogonal experiment could also vary the reward shown at the gate, which may matter more than the gate's position.

Methodology in Context

This analysis mirrors the structure of a production A/B-test readout — assignment balance check, primary metric significance test, secondary metric direction check, segment decomposition, and a robustness check — implemented end-to-end in SQL with no external statistical libraries. The same pattern generalizes directly to product experiments in subscription, e-commerce, payments, or fintech contexts: every step here (assignment table, outcome metric joins, conditional aggregation for arms, z-test math, segment cuts, bootstrap resampling) maps to a real experimentation pipeline.